D Bhanu Prakash

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Profile —

Dr. D Bhanu Prakash is an Assistant Professor of Mathematics in the Department of Mathematics and Computer Science at Sri Sathya Sai Institute of Higher Learning, India. He holds a Ph.D. in Mathematics, with research interests in Dynamical Systems and Optimal Control Theory, particularly as applied to Mathematical Biology. Prior to this appointment, He worked as an Assistant Professor at Vignan Deemed to be University. During his PhD, he served for three years as a Research Fellow on a DAE-NBHM funded project, gaining extensive experience in both theoretical and applied mathematical research. He has to his credit 8 publications and has served as a resource person in 6 workshops and FDPs, besides presenting at 3 conferences. This background has shaped a strong analytical foundation alongside a sustained commitment to teaching and mentoring students in mathematics.

Areas of Expertise

• Mathematical Modeling • Nonlinear Dynamical Systems • Optimal Control Theory

Experience

Assistant Professor **July 2026 – Ongoing**
Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning, India.

- I am teaching Optimization Techniques for Machine Learning, Data Visualisation using Python, and Probability and Statistics to BS Mathematics, Statistics and Computer Science students. My lecture material can be found in the link: https://dbhanuprakash233.github.io/SSSIHL_DBP/

Assistant Professor **June 2025 – June 2026**
Department of Mathematics and Statistics, School of Applied Science & Humanities, Vignan's Foundation for Science, Technology & Research (Deemed to be University), Guntur, India.

- I taught Fundamentals of Engineering Mathematics, Linear Algebra and Calculus and Ordinary Differential Equations to BTech CSE and AI students. My lecture material can be found in the link: https://dbhanuprakash233.github.io/CODE/ and /LA/ and /FEM/
- I served as an AI Champion of Vignan's flagship program, Agentic Vignan — the university's bold push to introduce and strengthen AI across campus.
- I am also a member of the Center of Excellence, Space Science and Technology.

Teaching Experience

Assistant Professor **July 2026 – Ongoing**
Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam - 515 134, India.

- Theory of Partial Differential Equations — Even Semester, 2026-2027
- Statistics for Data Science — Even Semester, 2026-2027
- Calculus of Variations — Even Semester, 2026-2027
- Mathematical Modeling — Even Semester, 2026-2027
- Optimization Techniques for Machine Learning — Odd Semester, 2026-2027
- Data Visualisation using Python — Odd Semester, 2026-2027
- Probability Theory and Distributions — Odd Semester, 2026-2027

- Bridge Mathematics - Value Added Course

My lecture material can be found in the link: https://dbhanuprakash233.github.io/SSSIHL_DBP/

Assistant Professor **June 2025 – June 2026**

Department of Mathematics and Statistics, Vignan's Foundation for Science, Technology & Research (Deemed to be University), Vadlamudi, Guntur - 522 213, India.

- 25MT101 - Fundamentals of Engineering Mathematics — Pre Semester, 2025-2026
- 25MT103 - Linear Algebra — Semester I, 2025-2026
- 25MT105 - Calculus and Ordinary Differential Equations — Semester II, 2025-2026

Teaching Assistant **Dec 2021 – Apr 2025**

Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning (SSSIHL) Prasanthi Nilayam - 515 134, India.

- Probability and Statistics — Semester II, 2021-2022
- Mathematical Modeling — Semester II, 2022-2023
- Calculus of Variations and Optimal Control Theory — Semester I, 2023-2024
- Real Analysis — Semester II, 2023-2024
- Linear Algebra — Semester I, 2024-2025
- Methods of Differential Equations — Semester II, 2024-2025

Education

PhD., Mathematics **Mar 2021 – Sept 2025**

Sri Sathya Sai Institute of Higher Learning (SSSIHL) *Prasanthi Nilayam - 515 134, India.*

- Thesis Title: "*Dynamics and Control of Additional Food Provided Prey-Predator Systems: A Deterministic and Stochastic Study with reference to Holling Type-III/IV Responses Incorporating Mutual Interference and Intraspecific Competition*".
- Research Supervisor: Dr. Krishna Kiran Vamsi Dasu
- Three journal papers were published and three papers were communicated. Delivered talks in two international and one national conferences.

M.Sc. Mathematics specialization in Computer Science **2018–2020**

Sri Sathya Sai Institute of Higher Learning (SSSIHL); GPA: 8.3/10 *Prasanthi Nilayam - 515 134, India.*

B.Sc. Mathematics (Hons.) specialization in Computer Science **2015–2018**

Sri Sathya Sai Institute of Higher Learning (SSSIHL); GPA: 7.5/10 *Prasanthi Nilayam - 515 134, India.*

Intermediate **2013–2015**

Sri Chaitanya Junior College; Score: 975/1000 *Machilipatnam - 521 001, India.*

S.S.C **2012–2013**

Sree Balajee Vidyalayam; GPA: 9.7/10 *Machilipatnam - 521 001, India.*

Certifications

Bayesian Statistics by Duke University offered through Coursera (Certificate) **Mar 2025**

Stochastic Processes by HSE University and offered through Coursera (Certificate) **Oct 2024**

Selected Publications

Preprints

- **D. B. Prakash** and D. K. K. Vamsi, "Deterministic and stochastic studies on additional food provided prey-predator systems with group defence among prey and mutual interference among predators," *arXiv preprint arXiv:2504.21114*, 2025. [Online]. Available: https://www.arxiv.org/abs/2504.21114 Manuscript submitted for review to *Journal of Analysis, Springer Nature*.

Journal Articles

- **D. B. Prakash**, C. Chaitanya, D. K. K. Vamsi, A. Aloqaily, and N. Mlaiki, "Prey-predator dynamics with additional food and holling type-iii response: Application to bemisia tabaci," *Scientific Reports*, vol. 16, no. 1, p. 25 718, 2026. [Online]. Available: https://doi.org/10.1038/s41598-026-58273-9
- **D. B. Prakash**, C. Chaitanya, and D. Vamsi, "Role of intra-specific competition and additional food on prey-predator systems exhibiting holling type-iv functional response," *Franklin Open*, p. 100 640, 2026. [Online]. Available: https://doi.org/10.1016/j.fraope.2026.100640
- **D. B. Prakash** and D. K. K. Vamsi, "Global dynamics and time-optimal control studies for additional food provided holling type-iii mutually interfering prey-predator systems with applications to pest management," *Discov Appl Sci 7, 874*, 2025. [Online]. Available: https://doi.org/10.1007/s42452-025-07255-z
- **D. B. Prakash** and D. K. K. Vamsi, "Stochastic time-optimal control and sensitivity studies for additional food provided prey-predator systems involving holling type-iv functional response," *Frontiers in Applied Mathematics and Statistics*, vol. 9, p. 1 122 107, 2023. [Online]. Available: https://doi.org/10.3389/fams.2023.1122107
- **D. B. Prakash** and D. K. K. Vamsi, "Stochastic optimal and time-optimal control studies for additional food provided prey–predator systems involving holling type iii functional response," *Computational and Mathematical Biophysics*, vol. 11, no. 1, p. 20 220 144, 2023. [Online]. Available: https://doi.org/10.1515/cmb-2022-0144
- B. Chhetri, V. M. Bhagat, D. K. K. Vamsi, V. S. Ananth, **D. B. Prakash**, S. Muthusamy, P. Deshmukh, and C. B. Sanjeevi, "Optimal drug regimen and combined drug therapy and its efficacy in the treatment of covid-19: A within-host modeling study," *Acta Biotheoretica*, vol. 70, no. 2, pp. 1–28, 2022. [Online]. Available: https://doi.org/10.1007/s10441-022-09440-8
- B. Chhetri, V. M. Bhagat, D. K. K. Vamsi, V. S. Ananth, **D. B. Prakash**, R. Mandale, S. Muthusamy, and C. B. Sanjeevi, "Within-host mathematical modeling on crucial inflammatory mediators and drug interventions in covid-19 identifies combination therapy to be most effective and optimal," *Alexandria Engineering Journal*, vol. 60, no. 2, pp. 2491–2512, 2021. [Online]. Available: https://doi.org/10.1016/j.aej.2020.12.011
- **D. B. Prakash**, D. K. K. Vamsi, D. B. Rajesh, and C. B. Sanjeevi, "Control intervention strategies for within-host, between-host and their efficacy in the treatment, spread of covid-19: A multi scale modeling approach," *Computational and Mathematical Biophysics*, vol. 8, no. 1, pp. 198–210, 2020. [Online]. Available: https://doi.org/10.1515/cmb-2020-0111

Workshop and Speaking Engagements

- **Resource Person**, FDP on *"Statistical Process Design using R & Python"*, Mangalore Institute of Technology & Engineering (MITE), Karnataka - May 8-10, 2025. Conducted a 4-hour lecture and hands-on session on **Stochastic Processes and Multivariate Analysis**.
- **Resource Person**, FDP on *"Advances in Mathematical Modeling Techniques, AI/ML & Bioinformatics"*, Sri Sathya Sai Institute of Higher Learning (SSSIHL) and Indian Society for Mathematical Modeling and Computer Simulation (ISMMACS) - April 12-13, 2025. Conducted a 2-hour hands-on session on **A Python primer on Optimal Control Problems**.
- **Resource Person**, Online workshop on *"Mathematical Modelling of Infectious Diseases"*, MedPro, Tamilnadu - February 20, 2025. Organized a two hour hands-on session on the topic: **Mathematical Modeling of Infectious Diseases using Python**.
- **Resource Person**, A 6-day In-person *"National Winter School for Women in AI and Computational Biology - 2024"*, Centre for Excellence in Mathematical Biology (CEMB-SSSIHL) - December 2-7, 2024. Organized a hands-on session for four hours on the topic: **Introduction to Programming and Disease Modeling using Python**. (Report)
- **Resource Person**, FDP on *"Dynamical Systems & Optimal Control Theory, AI/ML, and Bioinformatics with Applications to Healthcare"*, Centre for Excellence in Mathematical Biology (CEMB-SSSIHL)

- July 8-15, 2024. Organised a hands-on session for six hours on the topic: **Dynamical Systems and Optimal Control Theory with Python.**
- **Guest Lecture**, A 2-day In-person workshop on *"Infectious Disease Modeling"*, ICMR-National Institute for Research in Tuberculosis (ICMR-NIRT), Department of Health, Chennai, Government of India - March 14-15, 2024. Organised a hands-on session for four hours on the topic: **Introduction to Python Programming and Exploring Basic Disease Models in Python**.

Conference Talks

- *Deterministic and Stochastic Time Optimal Control Studies of Coupled Nonlinear Dynamical Systems with Applications to Pest Management.* **2nd International Conference on Recent Advances in Applied Mathematics** (RAAM 2024) organized by Indian Institute of Technology IIT BHU, Varanasi during July 2024.
- *Deterministic and Stochastic Studies on Additional Food Provided Prey-Predator System involving Holling Type-III and Holling Type-IV Functional Responses.* **National Conference on Recent Trends in Mathematical Biology - Theory, Methods and Applications** organized by Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning (SSSIHL) during July 2023.
- *Stochastic Time-Optimal Control Studies for Additional Food Provided Prey-Predator System involving Holling Type-IV Functional Response and Mutually Interfering Predators.* **International Conference on Differential Equations and Control Problems (ICDECP23)** organized by School of Mathematics and Statistical Sciences, Indian Institute of Technology Mandi (IIT Mandi) during June 2023.

Projects

Neural Differential Equations Framework **Link - Dec 2025**

- Implemented Neural ODEs, SDEs, and PINNs with memory-efficient adjoint backpropagation.
- Developed adaptive RK45 solver and Euler-Maruyama integrator for continuous-time learning.
- Applied to spiral dynamics, heat equation solving, and irregular time-series prediction.
- Achieved O(1) memory complexity for arbitrary-depth networks using adjoint sensitivity analysis, enabling training on resource-constrained hardware.

Workshops/Conferences Participation

- **Colloquium on Artificial Intelligence and Cybersecurity** (CAICS) organised in **offline mode** by the Indian Institute of Science (IISc), Bengaluru and Argonne National Laboratory, Department of State, USA during 31 August - 2 September, 2026.
- 12-Day Online Refresher Course on **Foundations of Data Science and Machine Learning for STEM Professionals** organized by NIT Warangal, Telangana during November 3–15, 2025.
- **14-day Faculty Development Program (FDP) on Foundations and Applications of Quantum Computing** in *online mode* by the Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning (SSSIHL), Andhra Pradesh, India during March 18- April 3, 2024.
- **Winter School on Games in Evolutionary Dynamics** organized in **completely offline mode** by Department of Mathematics, Shiv Nadar Institute of Eminence (Deemed to be University), Delhi NCR during December 18-23, 2023.
- **National Center for Mathematics Workshop on Control Theory for Partial Differential Equations (NCMW-CTPDE)** organized in **completely offline mode** by IISER, Thiruvananthapuram during December 04-16, 2023.
- **SERB Sponsored High-End Workshop (KARYASHALA) on Bifurcations and Chaos: Computations and Applications** organized in **completely offline mode** by Department of Mathematics, Indian Institute of Technology Indore (IIT Indore) during July 03-09, 2023.

Membership

- Graduate Student Member - Society for Industrial and Applied Mathematics (SIAM)
- Life Member - Indian Academy of Mathematical Modeling and Simulation (IAMMS)
- Life Member - Forum for Industrial and Applied Mathematics (FIAM)

Honors and Awards

- APSET 2021 - **Qualified** in Mathematics
- AP EAMCET 2015 - **Rank 1939**
- TS EAMCET 2015 - **Rank 2048**
- IIT JEE Main 2015 - **Rank 17003**

Technical Skills

Programming	Python, R, MATLAB, C, C++
Softwares	LaTeX, MS Office
OS	Ubuntu, macOS, Windows
Framework	PyTorch

Leadership and Extracurricular Activities

- Lead the team of Audio Control group at the University and Hostel during 2016-2018.
- Lead the team of the Hostel General Stores during 2019-2023.
- Participated in Sri Sathya Sai Village Service Program - Grama Seva, providing food and clothing to over 180 villages in Andhra Pradesh, India.
- An active volunteer in Service programs of Sri Sathya Sai Seva Organisations, India.

Personal Data

- Date of Birth: 23 March, 1998
- Nationality: Indian
- Sex: Male
- Marital Status: Single
- Languages: English, Telugu

References

- Dr. Krishna Kiran Vamsi Dasu (Ph.D. Supervisor), Associate Professor & Head of the Department, Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam - 515 134, Andhra Pradesh, India. dkkvamsi@sssihl.edu.in
- Prof. N Uday Kiran, Professor, Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam - 515 134, Andhra Pradesh, India. nudaykiran@sssihl.edu.in
- Prof. N Seshagiri Rao, Professor, Department of Mathematics, School of Physical Sciences, GMRIT (Deemed to be University), Rajam, Vizianagaram, Andhra Pradesh - 532 127, India. seshu.namana@gmail.com
- Prof. Pallav Kumar Baruah, Senior Professor & Dean of Sciences, Department of Mathematics and Computer Science, Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam - 515 134, Andhra Pradesh, India. pkbaruah@sssihl.edu.in